

Math 36 Final Project

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1 Introduction

I chose to analyze the 2024 and 2026 congressional district plans for the state of Louisiana. Louisiana’s district maps have undergone years of debate and change, culminating in recent a Supreme Court case with nationwide implications for redistricting. Louisiana is also a good state to study the impact of redistricting on minority vote representation given its large population of Black voters.

1.1 Background

Under state and federal law, Louisiana’s congressional and local districts must be redrawn every ten years, after each national census. Redistricting is controlled by the state legislature rather than an independent commission.¹ Louisiana typically elects Republicans to 7 out of 8 national congressional seats.² However, voter registration by party is approximately equal between Republicans and Democrats.³ Voting is divided across demographic lines; over 90% of Louisiana Republicans are white, while 90% of Black voters vote Democratic.⁴

1.2 History

Congress passed the Voting Rights Act in 1965, intended to prevent racial discrimination in registration and voting, especially in southern states like Louisiana. Section 2 of the VRA specifically prohibits discrimination in registration or voting based on race, color, or membership in a minority group.⁵

In 2022, the Louisiana legislature created a new congressional map based on the 2020 census. Plaintiffs, including individual voters and voting rights organizations such as the ACLU and Legal Defense Fund, challenged this map in court. In the lawsuit *Robinson v. Landry*, plaintiffs claimed the 2022 map violated Section 2 of the VRA by diluting the voting power of Black Louisianans and failing to give Black voters equal opportunity to elect representatives. Despite the fact that Louisiana’s population is approximately one-third Black, the 2022 congressional map created only majority-Black district out of six. This singular majority-Black district “packed” Black voters into one bloc.

After a long stretch of litigation, in November 2023, the Fifth Circuit Court of Appeals declared the plaintiffs in *Robinson v. Landry* were “likely to succeed” in proving that the 2022 map violated the VRA.⁶ In response to the court’s signal, on January 19th, 2024, the Louisiana Legislature and Governor Landry passed a new map with a second majority-Black congressional district. This map remained in place for the 2024 congressional and presidential elections, making 2024 the first time in a decade that Louisiana elected two Democrats to the House of Representatives.⁷

¹State of Louisiana, *Redistricting State of Louisiana*, Accessed June 5, 2026, redist.legis.la.gov, <https://redist.legis.la.gov/>.

²270 to Win, *Louisiana Presidential Election Voting History*, Accessed June 5, 2026, 2026, <https://www.270towin.com/states/louisiana>.

³Independent Voter Project, *Louisiana Voter Statistics*, Accessed June 5, 2026, 2026, <https://independentvoterproject.org/voter-stats/la>.

⁴Andrea Hagan, “The 6-0 Map Nobody Passed Tells You Everything About the 5-1 Map They Did”, Accessed June 5, 2026, *Louisiana Illuminator*, June 2026, <https://lailluminator.com/2026/06/02/6-0-map/>.

⁵Library of Congress, *The Voting Rights Act: Historical Development and Policy Background*, Accessed June 5, 2026, Congress.gov, 2025, <https://www.congress.gov/crs-product/R47520>.

⁶American Civil Liberties Union, *Robinson v. Landry*, Accessed June 5, 2026, American Civil Liberties Union, <https://www.aclu.org/cases/robinson-v-ardoin>; Mariel Fernandez, *Robinson v. Landry: Challenging Louisiana’s Congressional Maps*, Accessed June 5, 2026, Legal Defense Fund, May 2022, <https://www.naacpldf.org/case-issue/robinson-v-landry-louisiana-discriminatory-redistricting/>.

⁷PBS News, *Democrat Cleo Fields Flips Louisiana Congressional District*, Accessed June 5, 2026, PBS News, Nov. 2024,

1.3 Relevance

In January 2024, a group of “non-African American” Louisiana voters challenged the new map in court, claiming it was racial gerrymandering against the 14th Amendment. This case eventually reached the Supreme Court as *Louisiana v. Callais*. In April 2026, the Supreme Court determined that purposefully creating a second majority-Black district was an unconstitutional racial gerrymander and struck down Louisiana’s 2024 congressional map.

The Supreme Court’s decision in *Louisiana v. Callais* greatly diminished the power of Section 2 of the VRA, which originally allowed (and sometimes required) creating majority-minority voting districts to ensure minority representation. Now, states cannot be required by the VRA to create majority-minority districts, thereby making it harder to challenge congressional maps that dilute minority voting power. In May 2026, the Louisiana legislature passed a new map eliminating one of the two original majority-Black districts. This new map is expected to produce a 5–1 Republican advantage in Louisiana’s congressional elections.⁸

2 Ensemble Analysis

2.1 Aim

The goal of this project is to use ensemble analysis to analyze Louisiana’s 2024 and 2026 congressional maps.

The idea behind ensemble analysis is to generate a large collection of district maps drawn without any demographic or partisan agenda, creating a “neutral” distribution of plans. Through ensemble analysis, we can compare how the 2024 and 2026 enacted plans fall on certain measures compared to the the distribution of these metrics among the randomly generated plans.

If the enacted maps’ scores fall within the random maps’ score distributions, the enacted maps are similar to a map that would have been generated without a goal (e.g. creating a majority-Black district) in mind. If the enacted plans fall outside or at the tails of the generated distributions, it would suggest the maps would not have been generated by a neutral scoring function.

2.2 Data

All data was sourced from the Redistricting Data Hub. First, I used Louisiana 2024 precinct-level data, including 2024 congressional election results and each precinct’s congressional district assignment.⁹ I used 2020 Census data for population and demographics numbers, and matched these figures onto each 2024

<https://www.pbs.org/newshour/politics/democrat-cleo-fields-flips-louisiana-congressional-district>.

⁸Jane Timm, *Louisiana passes new congressional map, dismantling one majority-Black district*, Accessed June 5, 2026, NBC News, May 2026, <https://www.nbcnews.com/politics/2026-election/louisiana-passes-new-congressional-map-dismantling-one-majorityblack-rcna347575>; Scott Bomboy, *The Supreme Court’s Callais decision sets new framework for racial gerrymandering*, Accessed June 5, 2026, National Constitution Center, Apr. 2026, <https://constitutioncenter.org/blog/the-supreme-courts-callais-decision-sets-new-framework-for-racial-gerrymandering>.

⁹Redistricting Data Hub, *Louisiana 2024 General Election Precinct-Level Results (November 2024)*, Accessed May 30, 2026, <https://redistrictingdatahub.org/dataset/louisiana-2024-general-election-precinct-level-results-november-2024/>; Redistricting Data Hub, *2024 Louisiana Congressional Districts Approved Plan*, Accessed May 30, 2026, <https://redistrictingdatahub.org/dataset/2024-louisiana-congressional-districts-approved-plan/>.

precinct.¹⁰ I also used the 2026 enacted map, passed just this week.¹¹ I assigned each precinct to its 2026 district to evaluate what would happen in the 2024 election under the new district lines and compare the projected 2026 election results to the ensemble.

2.2.1 Population, Demographic and District Maps

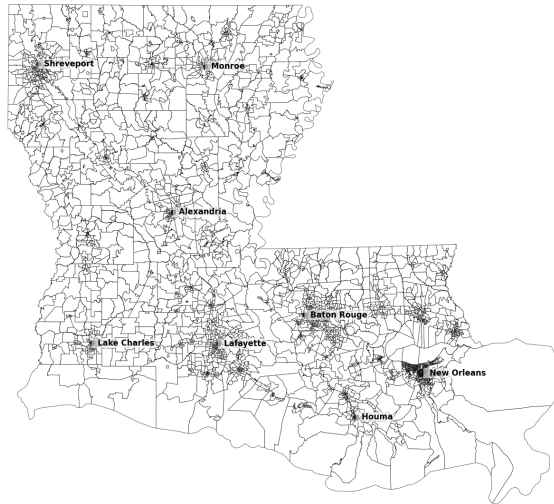


Figure 1: Louisiana Precincts, Cities (2024)

Note: 2024 precincts are used as the base for all plans and maps, assuming precincts remain stable enough for comparison.

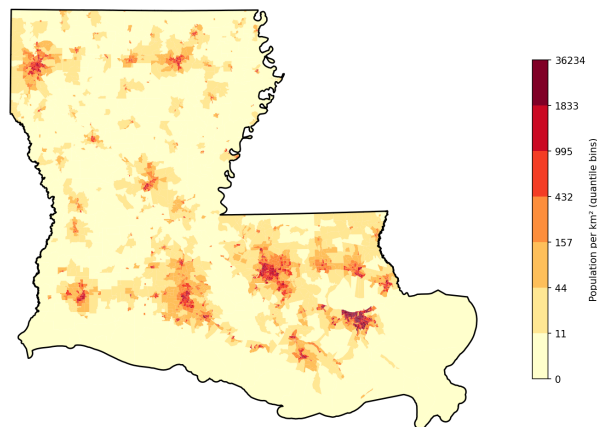


Figure 2: Louisiana Population Density (2020 Census)

¹⁰Redistricting Data Hub, *Louisiana VTD PL 94-171 2020 Official*, Accessed May 30, 2026, <https://redistrictingdatahub.org/dataset/louisiana-vtd-pl-94-171-2020-official/>; Redistricting Data Hub, *Louisiana Block PL 94-171 2020*, Accessed May 30, 2026, <https://redistrictingdatahub.org/dataset/louisiana-block-pl-94171-2020/>.

¹¹Redistricting Data Hub, *2026 Louisiana Congressional Districts Approved Plan*, Accessed June 3, 2026, <https://redistrictingdatahub.org/dataset/2026-louisiana-congressional-districts-approved-plan/>.

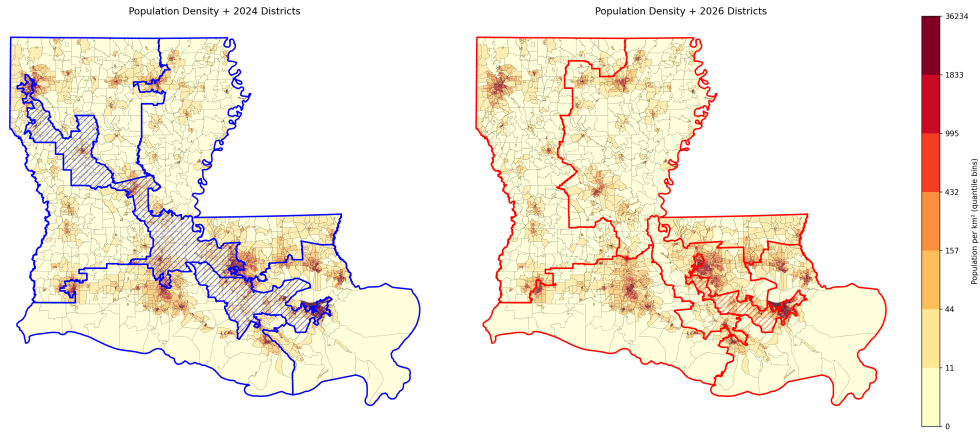


Figure 3: Louisiana Population Density and District Maps
Shaded districts are majority-Black population districts.

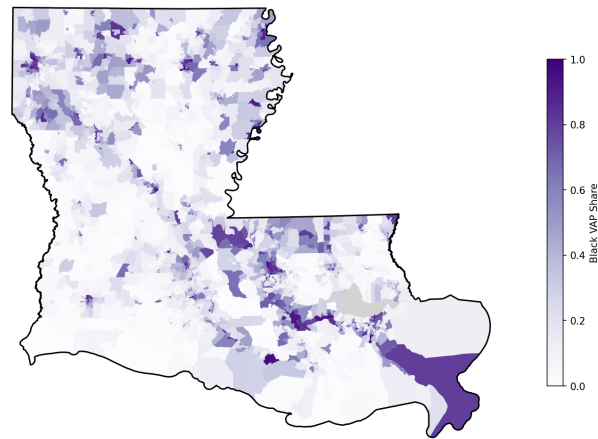


Figure 4: Louisiana Black Voting Age Population Share (2020 Census)

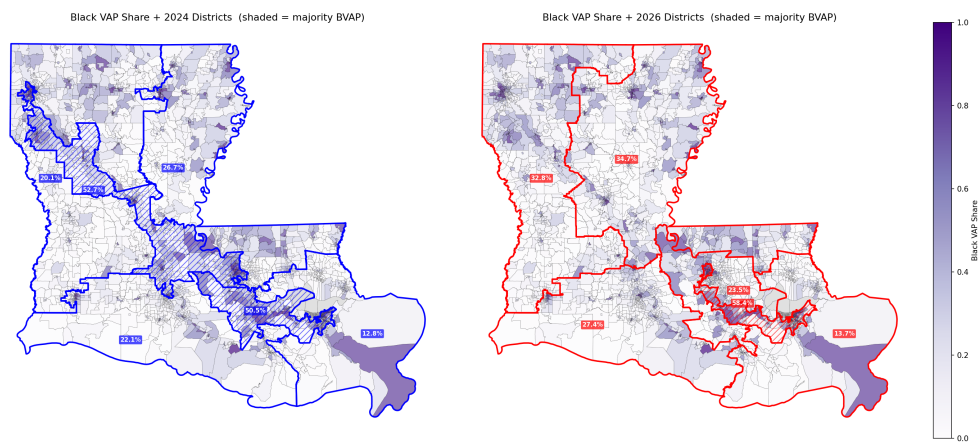


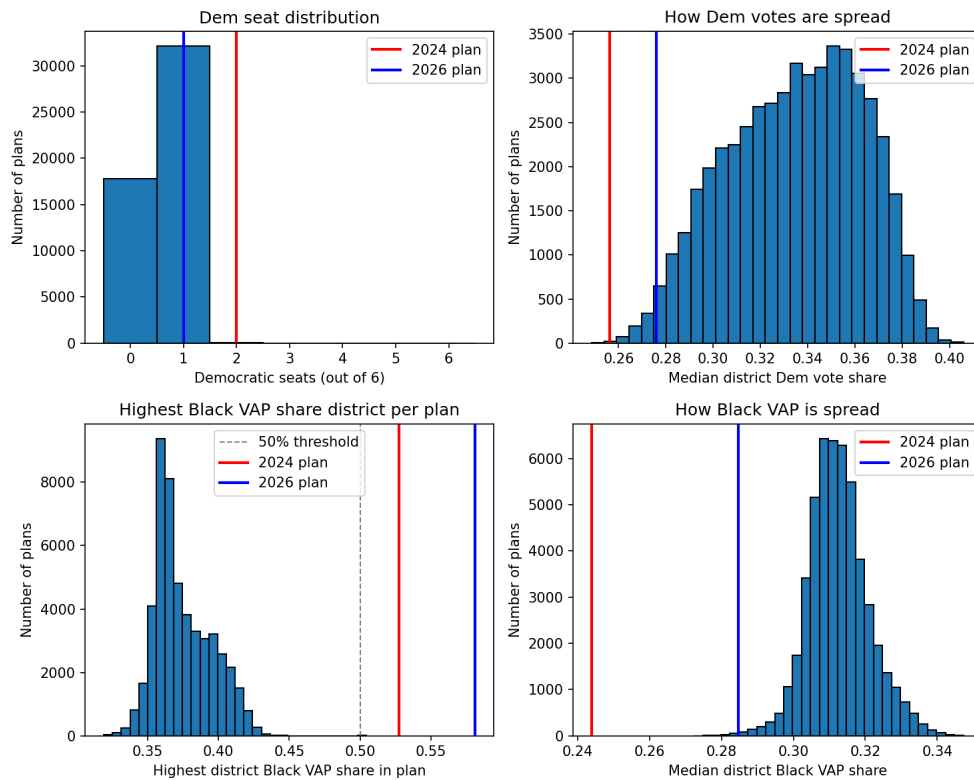
Figure 5: Louisiana Black Voting Age Population Share and District Maps

To run the ensemble analysis, I used the Metric Geometry and Gerrymandering Group’s GerryChain library.¹² The analysis uses Markov Chain Monte Carlo methods. Starting from the 2024 enacted map, I used GerryChain’s recombination algorithm to generate a sample of new maps. Recombination merges two neighboring districts, draws a spanning tree across the combined region, and cuts one edge of that tree to split it back into two new districts. The algorithm only proposes new maps where each district falls within 5% of equal population, so population balance is built into how the maps are generated.

The recombination algorithm does not move towards any particular kind of map regarding party or demographic distributions. Its only tendency is towards compact districts, which is a function of the spanning-tree step. The resulting distribution is therefore not uniform over all possible plans, but neutral with respect to the features of interest in most redistricting efforts (e.g. partisanship or race).

I generated 50,000 maps and used 2024 election and 2020 census data to calculate four metrics for each: Democratic seat count, spread of Democratic votes, highest Black Voting Age Population (VAP) share in a district, and median district Black VAP share. The result is a distribution and expected range for these outcomes across the 50,000 “neutrally” drawn maps. Lastly, I placed the 2024 and 2026 enacted plans on these distributions to see where they landed relative to the randomly generated maps.

2.3 Results



As seen in these figures, the 2024 and 2026 enacted plans both fall outside or near the edge of the distributions of metrics measured from randomly generated plans.

¹²Metric Geometry and Gerrymandering Group, *GerryChain*, Accessed May 30, 2026, <https://github.com/mggg/GerryChain>.

Of the 50,000 generated plans, just over 30,000 yield one Democratic seat and around 17,500 yield none (given 2024 congressional election results). The 2026 plan falls at the median with one Democratic seat, but the 2024 plan is a clear outlier with two seats.

Looking at the spread of Democratic votes, the median district's Democratic vote share from the generated maps is concentrated from 30-38%. This finding aligns with 2024 congressional election results, where approximately 34.5% of Louisiana votes were Democratic.¹³ Meanwhile, the 2026 plan yields a median district Democratic vote share of around 0.275. 2024's is even lower at approximately 0.25. These lower median district Democratic vote shares indicate that Democratic votes were concentrated in a few districts, leaving the remaining districts (including the median) with lower Democratic vote shares.

These figures show similar trends for district Black VAP share. The randomly generated maps yielded a distribution of highest district Black VAP share with a median around 0.37 and a distribution of median district Black VAP share with a median around 0.31. These numbers follow from the fact that Louisiana's population is approximately one-third Black, so Black VAP would be around 33% in randomly generated districts with some variability given population concentration in certain areas.

Because of majority-minority district(s), the highest district Black VAP shares in the 2024 and 2026 plans are over 50%. In the 2024 plan, this highest district share is a little lower than in the 2026 plan since the 2024 plan needed to have two majority-Black districts, giving each a slim majority. Median district Black VAP share in the enacted plans is lower than in the ensemble maps; creating majority-Black districts means that the other districts (including the median district) must have lower proportions of Black voters.

All in all, it is clear that the majority-minority district(s) in the 2024 and 2026 plans are highly unlikely to arise from randomly generated maps via recombination. This being said, almost two-thirds of plans created by recombination yielded one Democratic seat, meaning that even if future congressional maps have no majority-minority districts, there is a high chance Louisiana retains one Democratic seat.

3 K-means Clustering

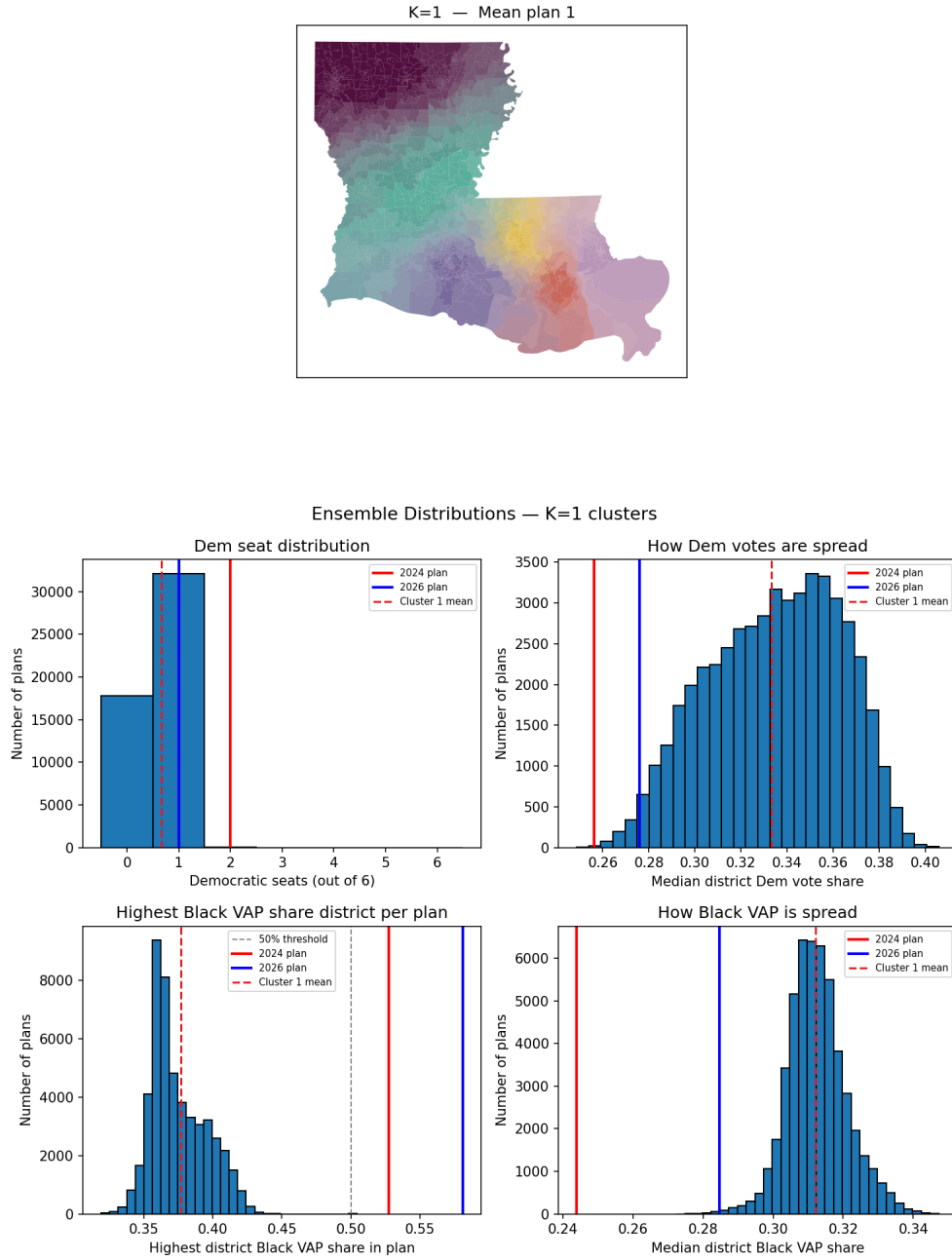
Finally, I used K-means clustering to group a random sample of 1,000 ensemble maps into a few types of district maps. I used Thomas Weighill's *minigerryclusters* python code, which sorts the maps into K groups such that maps within a group resemble one another, i.e. they divide Louisiana's precincts into districts in similar ways.¹⁴ Using this algorithm, I generated ice cream plots visualizing each map group for $K = 1, 2, 4$, and 8.

The point of this exercise was to see whether any map groups could produce districts that group minority voters in a similar way to the 2024 congressional map without an explicit instruction to create majority-Black districts. To see this clearly, I calculated the average of each of the four metrics across the maps in each cluster and plotted these cluster averages against the ensemble distributions and enacted plans.

¹³Wikipedia, *2024 United States House of Representatives elections in Louisiana*, Accessed June 5, 2026, Wikipedia, June 2026, https://en.wikipedia.org/wiki/2024_United_States_House_of_Representatives_elections_in_Louisiana.

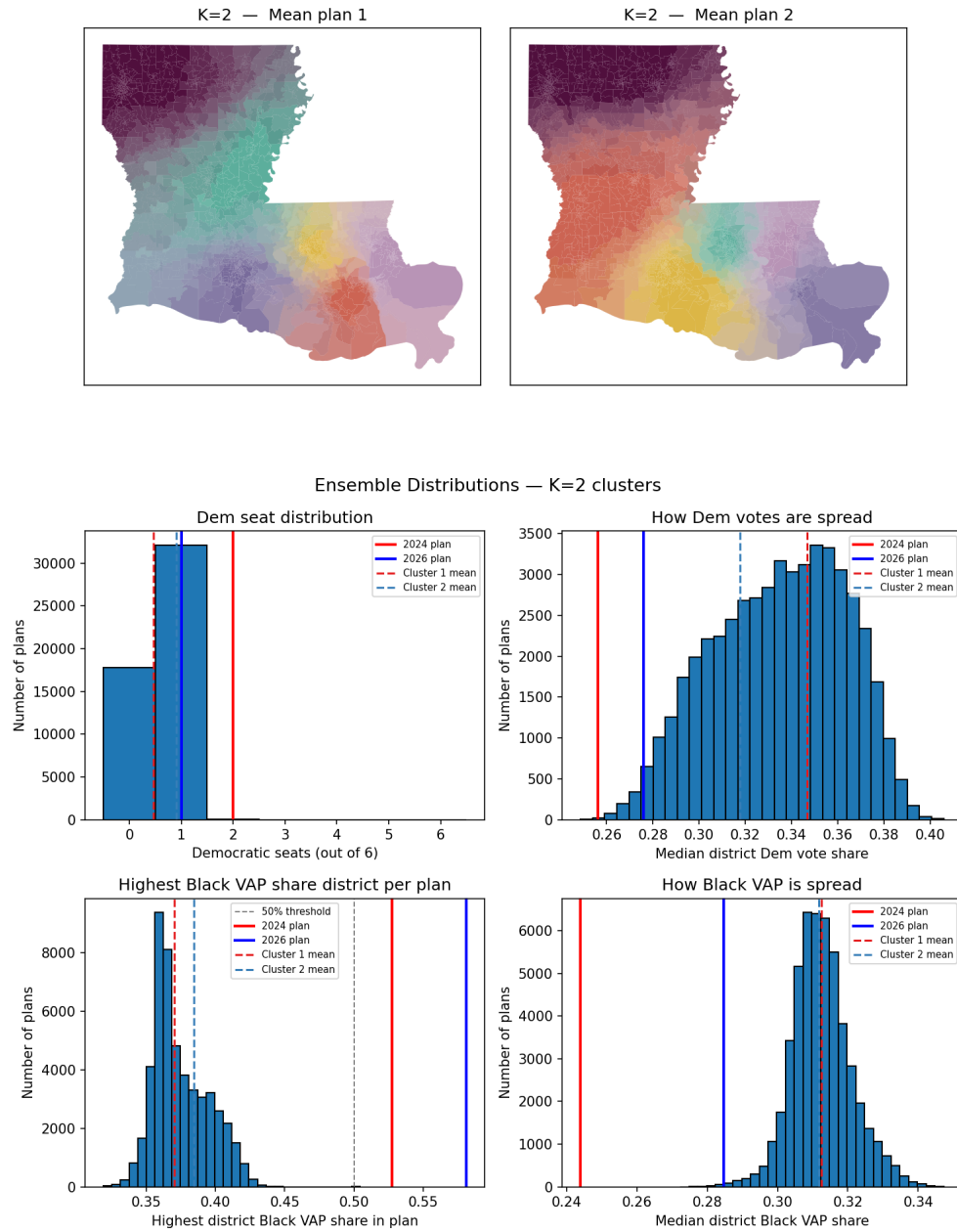
¹⁴Thomas Weighill, *minigerryclusters*, Accessed June 3, 2026, <https://github.com/thomasweighill/minigerryclusters>.

3.1 $K = 1$

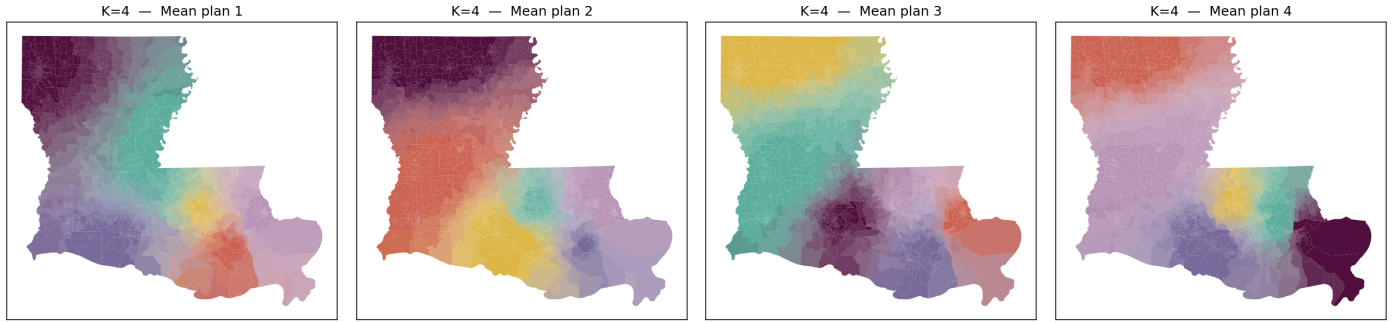


When $K=1$, there is a single group containing all sampled maps, so it follows that the cluster's average metrics fall near the median of the ensemble distributions.

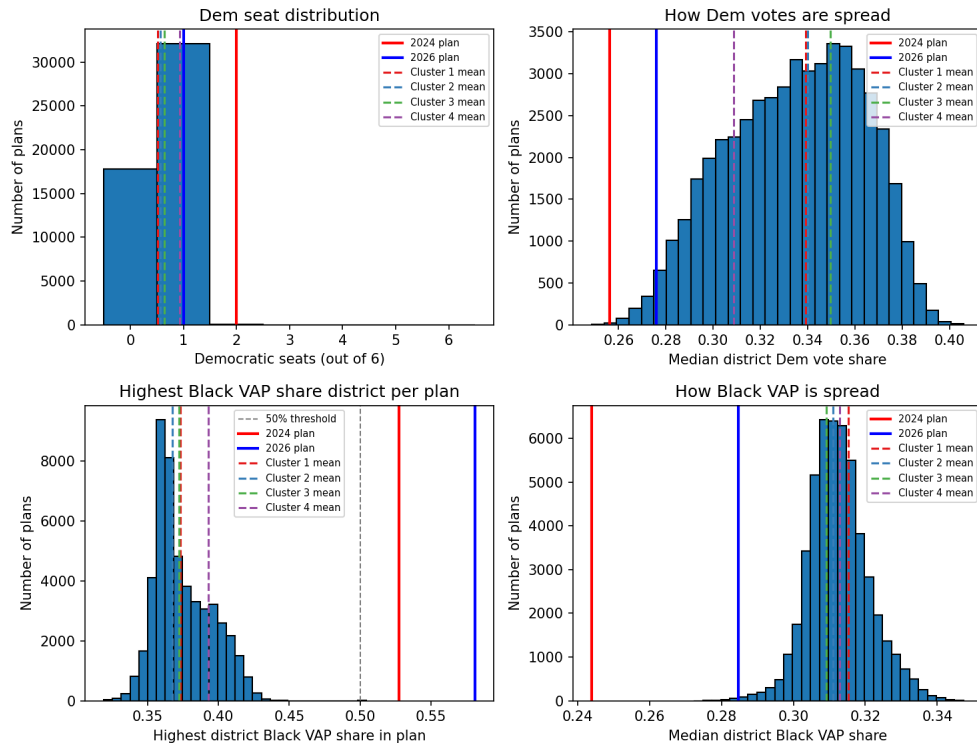
3.2 $K = 2$



3.3 $K = 4$

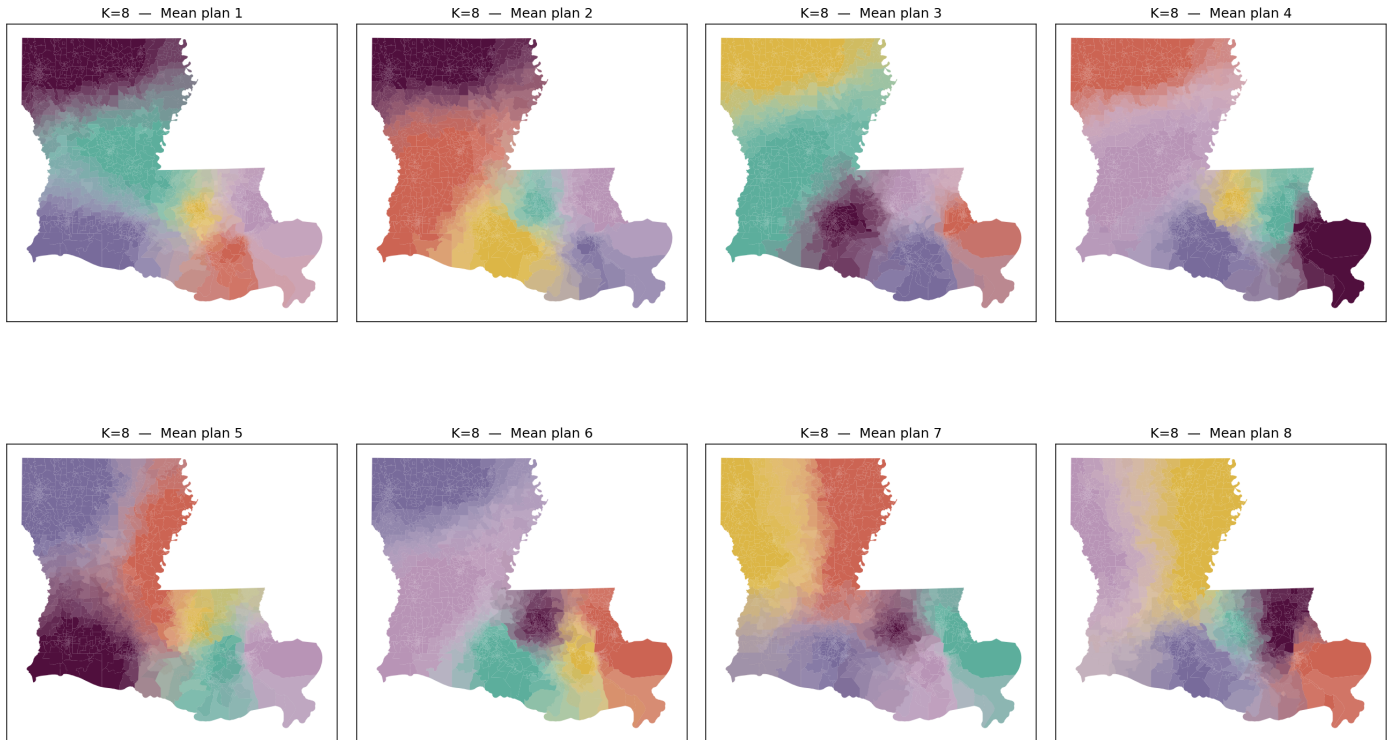


Ensemble Distributions — K=4 clusters

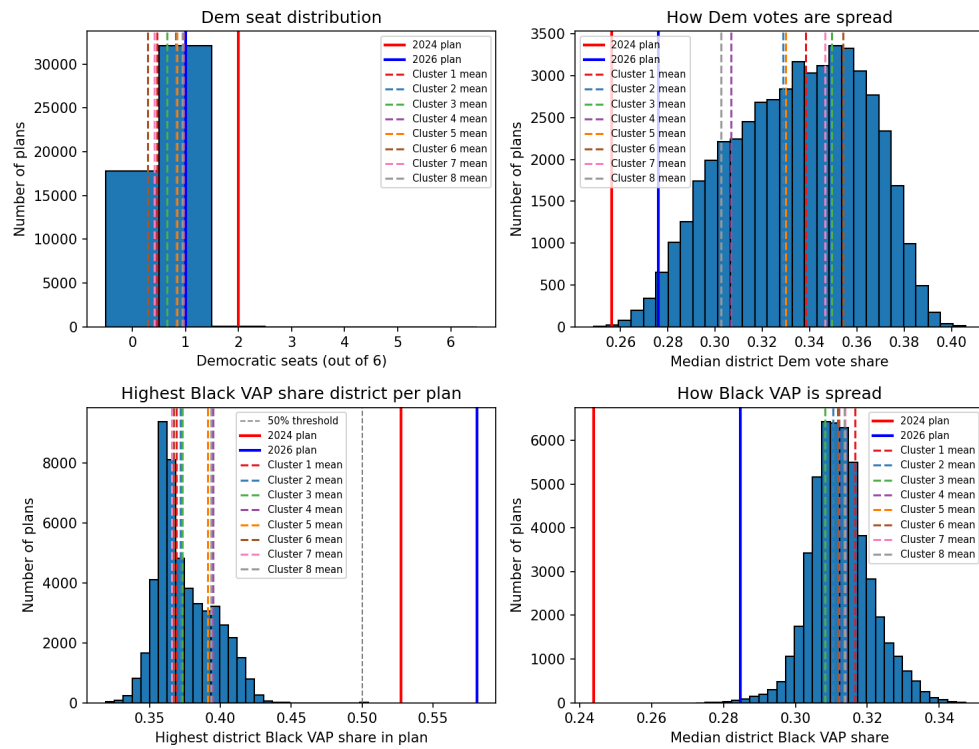


Using 2 or 4 groups introduces more variation, since the average metrics are calculated separately for each cluster. Still, even with 2 or 4 averages, the cluster mean metrics still fall near the center of the ensemble distributions.

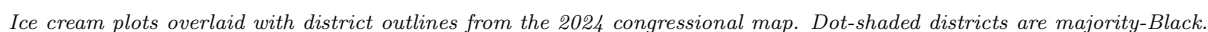
3.4 $K = 8$



Ensemble Distributions — $K=8$ clusters



Still, all other metrics indicate that even when maps are organized into 8 groups, the groups hold maps with similar average district partisan and demographic makeup. Considering that the K-means clustering process uses a random sample of graphs from the ensemble analysis, and very few of the ensemble maps had similar metrics to the enacted plans, it follows that none of the K-means graphs have districts with outlying demographic or partisan makeup.



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4 Implications and Next Steps

Using this method of ensemble analysis to evaluate the 2024 and 2026 congressional maps is a limited approach. The recombination algorithm creates districts without regard to many common districting considerations, such as preserving community structure or ensuring minority representation. The only guidelines ReCom follows are maintaining roughly equal district population and, as a function of the spanning-tree method, creating relatively compact districts.

Applying a scoring function to measure how maps perform on these common considerations, then using a Metropolis-Hastings accept/reject rule based on those scores, would create an ensemble of maps that satisfy a chosen set of criteria. For example, advocates of the 2024 congressional map justified its second majority-Black district, which spanned diagonally from Northwest to central Louisiana (Shreveport to Baton Rouge), on the grounds that it followed the Red River Valley. According to an amicus brief filed in response to the Louisiana redistricting cases, “The Red River Valley has been recognized as a cohesive geographical, agricultural, economic, social, and political unit for over two hundred years.”¹⁵ If future maps are optimized for cultural, geographical, or historical cohesion, maps with electoral outcomes similar to the 2024 map could be created without an explicit mandate for a majority-Black district.

One way to do this is to consider Communities of Interest when redistricting. COIs are geographic areas defined by a group of residents with significant shared interests that warrant needing joint representation in government. MGGG and other organizations are working on making COI data collection easier, and more jurisdictions have been considering COIs in the redistricting process.¹⁶

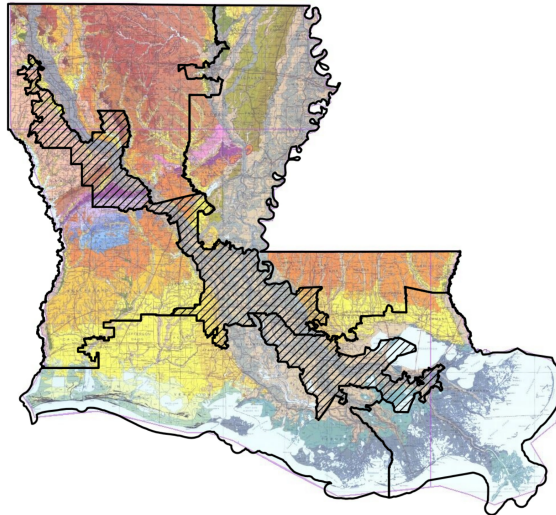


Figure 6: Geologic map of Louisiana overlaid with 2024 congressional district outlines. Adapted from the Louisiana Historians amicus brief (2024).

¹⁵Southern Coalition for Social Justice, *Louisiana Historians Urge U.S. Supreme Court to Allow Congressional Map with Second Majority-Black District*, Accessed June 5, 2026, Dec. 2024, <https://southerncoalition.org/louisiana-historians-urge-u-s-supreme-court-to-allow-congressional-map-second-majority-black-district/>.

¹⁶Metric Geometry and Gerrymandering Group, *Participatory Mapping and Communities of Interest*, Accessed June 5, 2026, 2021, <https://mggg.org/cois>.

References

- 270 to Win. *Louisiana Presidential Election Voting History*. Accessed June 5, 2026, 2026. <https://www.270towin.com/states/louisiana>.
- American Civil Liberties Union. *Robinson v. Landry*. Accessed June 5, 2026. American Civil Liberties Union. <https://www.aclu.org/cases/robinson-v-ardoin>.
- Bomboy, Scott. *The Supreme Court's Callais decision sets new framework for racial gerrymandering*. Accessed June 5, 2026. National Constitution Center, Apr. 2026. <https://constitutioncenter.org/blog/the-supreme-courts-callais-decision-sets-new-framework-for-racial-gerrymandering>.
- Fernandez, Mariel. *Robinson v. Landry: Challenging Louisiana's Congressional Maps*. Accessed June 5, 2026. Legal Defense Fund, May 2022. <https://www.naacpldf.org/case-issue/robinson-v-landry-louisiana-discriminatory-redistricting/>.
- Hagan, Andrea. "The 6-0 Map Nobody Passed Tells You Everything About the 5-1 Map They Did". Accessed June 5, 2026, *Louisiana Illuminator*, June 2026. <https://lailluminator.com/2026/06/02/6-0-map/>.
- Independent Voter Project. *Louisiana Voter Statistics*. Accessed June 5, 2026, 2026. <https://independentvoterproject.org/voter-stats/la>.
- Library of Congress. *The Voting Rights Act: Historical Development and Policy Background*. Accessed June 5, 2026. Congress.gov, 2025. <https://www.congress.gov/crs-product/R47520>.
- Metric Geometry and Gerrymandering Group. *GerryChain*. Accessed May 30, 2026. <https://github.com/mggg/GerryChain>.
- . *Participatory Mapping and Communities of Interest*. Accessed June 5, 2026, 2021. <https://mggg.org/cois>.
- PBS News. *Democrat Cleo Fields Flips Louisiana Congressional District*. Accessed June 5, 2026. PBS News, Nov. 2024. <https://www.pbs.org/newshour/politics/democrat-cleo-fields-flips-louisiana-congressional-district>.
- Redistricting Data Hub. *2024 Louisiana Congressional Districts Approved Plan*. Accessed May 30, 2026. <https://redistrictingdatahub.org/dataset/2024-louisiana-congressional-districts-approved-plan/>.
- . *2026 Louisiana Congressional Districts Approved Plan*. Accessed June 3, 2026. <https://redistrictingdatahub.org/dataset/2026-louisiana-congressional-districts-approved-plan/>.
 - . *Louisiana 2024 General Election Precinct-Level Results (November 2024)*. Accessed May 30, 2026. <https://redistrictingdatahub.org/dataset/louisiana-2024-general-election-precinct-level-results-november-2024/>.
 - . *Louisiana Block PL 94-171 2020*. Accessed May 30, 2026. <https://redistrictingdatahub.org/dataset/louisiana-block-pl-94171-2020/>.
 - . *Louisiana VTD PL 94-171 2020 Official*. Accessed May 30, 2026. <https://redistrictingdatahub.org/dataset/louisiana-vtd-pl-94-171-2020-official/>.
- Southern Coalition for Social Justice. *Louisiana Historians Urge U.S. Supreme Court to Allow Congressional Map with Second Majority-Black District*. Accessed June 5, 2026, Dec. 2024. <https://southerncoalition.org/louisiana-historians-urge-u-s-supreme-court-to-allow-congressional-map-second-majority-black-district/>.
- State of Louisiana. *Redistricting State of Louisiana*. Accessed June 5, 2026. redist.legis.la.gov. <https://redist.legis.la.gov/>.
- Timm, Jane. *Louisiana passes new congressional map, dismantling one majority-Black district*. Accessed June 5, 2026. NBC News, May 2026. <https://www.nbcnews.com/politics/2026-election/louisiana-passes-new-congressional-map-dismantling-one-majorityblack-rcna347575>.
- Weighill, Thomas. *minigerryclusters*. Accessed June 3, 2026. <https://github.com/thomasweighill/minigerryclusters>.
- Wikipedia. *2024 United States House of Representatives elections in Louisiana*. Accessed June 5, 2026. Wikipedia, June 2026. https://en.wikipedia.org/wiki/2024_United_States_House_of_Representatives_elections_in_Louisiana.